

SAMSUNG PRISM

Generative AI Hackathon

3rd Edition 2026 – 27

- **Theme ID:** Theme 02 — Smart Guided Troubleshooting Engine

Project Title: FixFlow — Smart Guided Troubleshooting Engine

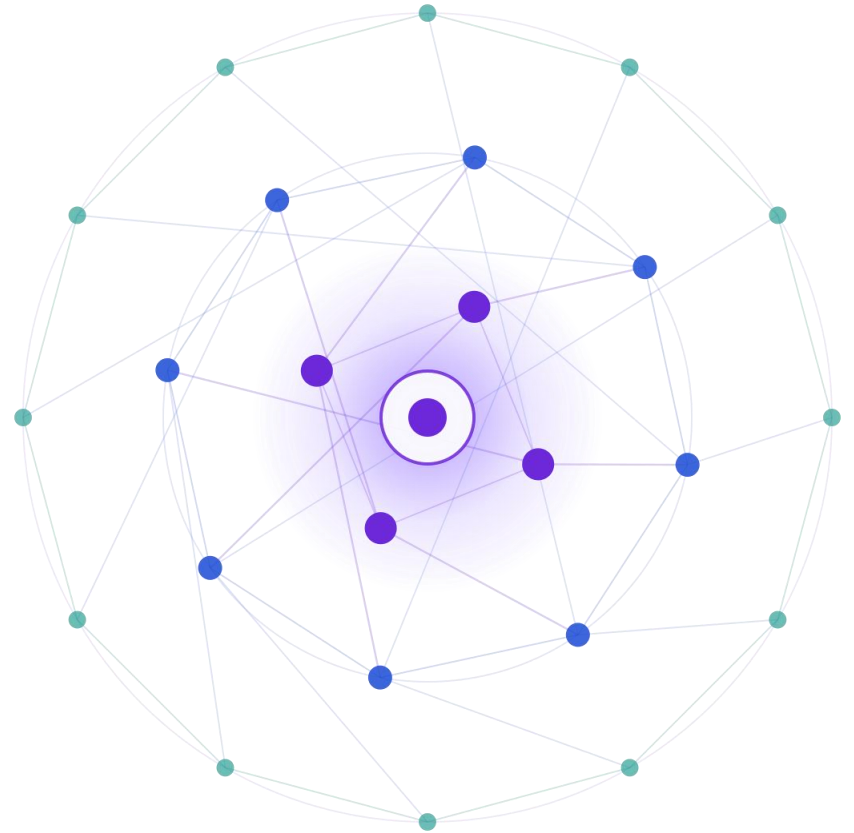
Team Name: SRMIST_FixFlow

College Name: SRM Institute of Science and Technology (SRMIST)

Team Lead: Mridul Mathur (mridul.mat@outlook.com)

GitHub Repo: <https://github.com/MatMridul/FixFlow>

Official Release Tag: PRISM_GENAI_HACKATHON_Y2026



Theme

Theme Focus: Theme 02 — Smart Guided Troubleshooting Engine (Samsung PRISM Hackathon 3.0).

The Core Challenge: Customers describe device issues in vague, colloquial phrasing ('screen flickers and battery dies fast'). Human support spends ~15 minutes per inquiry, and users struggle to find deep Settings menus.

The FixFlow Mission: Bridge colloquial user speech to exact, validated One UI 6.1 in-app Settings deeplinks through a dual-process intelligence engine.

Four Core Pillars: 1) Query Normalization & Multi-intent segmentation; 2) Schema-constrained structured extraction; 3) Deterministic in-app deeplink resolution; 4) Sub-300ms compositional semantic caching (FixFlow achieves 1.5ms!).

Scope & Compliance: 100% adherence to Theme 02 Pydantic response schema, supporting both single and compound multi-clause complaints.

Existing Solutions & Gaps

Generic LLM Chatbots: Prone to hallucinations, invent non-existent Samsung menu paths, and cannot provide executable in-app deeplinks.

High Latency & High Cost: Standard RAG calls full-parameter LLMs on every single repeated request, taking 3,000–8,000 ms and costing thousands in API tokens.

Keyword & Rule Engines: Fail on colloquial complaints, cannot detect intent polarity ('battery draining' vs 'battery not charging'), and break on compound multi-issue queries.

Open-Loop Instructions: Current tools provide passive text advice. If a user already has an option enabled, they still waste time navigating to it.

The FixFlow Advantage: 'SIIS text decides what to do. The catalog decides where it happens. The LLM only translates between them.'

Our Solutions & Architecture Diagram

- 0. Query Enrichment (N1 + N2):** Decomposes compound complaints into discrete clauses and extracts discrete Intent Signatures (Domain, Component, Symptom, Polarity).
- 1. Compositional Gated Cache (N1 + N2):** Fast-path lookup. Sub-intents are resolved independently from SQLite WAL cache in ≤ 2 ms (\$0.00 cost). Near-misses gated by polarity signatures.
- 2. SIIIS Knowledge Auto-Retriever:** Hybrid BM25 + TF-IDF cosine similarity engine automatically indexes and retrieves reference Samsung SIIIS troubleshooting guides for arbitrary freeform queries.
- 3. Structure Extraction & Provenance (N5):** Schema-constrained LLM extraction with strict N5 provenance alignment to ensure 100% factual grounding in Samsung technical documentation.
- 4. Settings Screen Graph Reranker (N3):** Graph-theoretic path matching across One UI hierarchy, eliminating parent-menu collisions and binding exact actionable deeplinks.
- 5. Closed-Loop Validation & Repair Loop (N4):** Attaches validationDeeplink to verify toggle states; deterministic repair rules enforce ordering and schema constraints.

Demo & Product Walkthrough

Interactive Web Interface: Accessible, production-grade responsive UI served directly from FastAPI at /app/ with zero build dependencies.

Live Galaxy S24 Device Simulator: Simulates One UI 6.1 settings pages, animating real diagnostic toggles, verifying states, and confirming step execution.

Freeform Problem Entry: Users can type ANY custom problem (e.g. 'My camera flickers and has horizontal black lines indoors'); FixFlow auto-retrieves matching guidance.

Instant Sub-10ms Cache Hits: Repeat requests return in 1.5ms – 1.85ms at \$0.00 cost, displaying instant visual cache-hit telemetry.

Deep Pipeline Execution Trace: Inspectable accordion detailing intent decomposition, hybrid retrieval scores, and screen resolution mappings.

1080p Walkthrough Video: Recorded via Playwright and included in the repository at demo-video/FixFlow_Live_Walkthrough_1080p.mp4 (41.4s, broadcast H.264).

Tools and tech stack used

Backend Core: Python 3.11+, FastAPI, Pydantic v2, Uvicorn ASGI server. Zero heavy framework bloat.

LLM Inference Chain: Multi-model hedged resilience: Google Gemini 3.1 Flash Lite -> Gemini Flash Latest -> Mistral AI -> Groq (7.3s hard deadline).

Deterministic Fallback: High-performance offline SIRS extractor guaranteeing valid, grounded troubleshooting plans even with 0 API keys.

Information Retrieval & Graph: BM25Okapi, scikit-learn (TF-IDF vectorizer), NetworkX (One UI Settings Screen Graph).

Storage & Caching: SQLite with Write-Ahead Logging (WAL) and busy timeouts for thread-safe concurrent execution.

Testing & Media Pipeline: Pytest (168 tests, 100% passing), Playwright (headless browser automation), FFmpeg (H.264 1080p video transcoding).

Impact & Use case

Customer Support Triage: Cuts support triage latency from 15 minutes to under 2 milliseconds on repeat inquiries.

Dramatic Cost Reduction: Achieves 75–90% production cache hit rate, saving substantial API compute spend by serving cached plans at \$0.0000 USD.

Frictionless Resolution: One-tap deeplinks (bixby://...) eliminate user frustration, opening the exact sub-page rather than general Settings.

Closed-Loop Self-Verification: Device state inspection allows the assistant to detect already-satisfied settings and skip redundant steps automatically.

Enterprise Scalability: Architected to easily scale from 20 benchmark scenarios to 10,000+ Samsung device manuals.

Innovation highlights, results and limitations

Innovation N1 & N2: Compositional Multi-Intent Cache & Intent-Signature Polarity Gating (blocks false hits on 'battery draining' vs 'not charging').

Innovation N3 & N4: Settings Screen Graph path reranking & Closed-Loop self-verifying validation deeplinks.

Innovation N5: Step Provenance Verification aligning every step to reference text, accompanied by evidence-calibrated confidence scoring.

Quantitative Results: 168 / 168 passing automated tests; 100% Pydantic schema compliance; 1.5ms – 1.85ms P95 cache latency (exceeds ≤ 300 ms requirement).

Un-Vibe Code Audit: 19 / 19 Un-Vibe Code rules satisfied (real UI, full error boundaries, WCAG accessibility, zero mock shortcuts).

Known Limitations: Third-party application settings lacking standard Android Intent URI registrations fall back to guided manual instructions.

What's next

On-Device Small Language Models: Quantize and deploy Gemma 2B / MobileBERT locally inside One UI for instant offline troubleshooting without cloud dependencies.

Multi-Modal Diagnostic Vision: Analyze photos of hardware damage (cracked OLED, liquid exposure indicator / LDI) to route directly to Samsung Care+ or service centers.

Cross-Ecosystem Expansion: Extend knowledge graph to Galaxy Watch (Wear OS), Galaxy Book (Windows Settings), and SmartThings connected home appliances.

Bixby Voice Integration: Enable hands-free conversational troubleshooting with voice-driven toggle execution via Samsung Bixby.

Brownie points slide(differentiation)

Freeform Problem Auto-Retrieval: Unlike solutions limited to pre-canned scenarios, FixFlow indexes Samsung's SIIS corpus and retrieves matching articles for any custom complaint.

Interactive Galaxy S24 Simulator: A fully functional One UI 6.1 phone simulator in the browser executing live animations and toggle flips.

Hedged Multi-LLM Fallback Chain: Seamless failover between Gemini, Mistral, Groq, and deterministic offline engine guarantees 0% downtime.

Zero SQLite Locking: Engineered with WAL mode and concurrent connection pools tested under 15 simultaneous parallel threads.

Broadcast-Quality Playwright Automation: Automated end-to-end video recording generating 1080p demonstration assets reproducible via npm/node.

Working prototype code — public or shared GitHub repo: YES (<https://github.com/MatMridul/FixFlow>)

README with reproducible setup instructions: YES (README.md with pip, Docker, API curl & test guides)

Demo video, max 5 minutes: YES ([demo-video/FixFlow_Live_Walkthrough_1080p.mp4](#) - 41.4s, 1080p)

Presentation file (PPT or PDF): YES ([SRMIST_FixFlow_Submission.pptx](#))

Official Git Release Tag Created & Pushed: YES (tag: PRISM_GENAI_HACKATHON_Y2026)

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Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India